

Phần 1 — pfSense: Network & Security

Cấu hình Cài đặt pfSense, cấu hình WAN (NAT ra Internet lab) và LAN 10.10.10.0/24

```
WAN (wan)      -> em0      -> v4/DHCP4: 192.168.232.161/24
LAN (lan)      -> em1      -> v4: 10.10.10.1/24
SERVER (opt1)  -> em1.10   -> v4: 10.10.20.1/24
CLIENT (opt2)  -> em1.20   -> v4: 10.10.30.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                    14) Enable Secure Shell (sshd)
6) Halt system                      15) Restore recent configuration
7) Ping host                        16) Restart PHP-FPM
8) Shell

Enter an option:
Message from syslogd@pfSense at Aug 24 02:03:17 ...
php-fpm[4421]: /index.php: Successful login for user 'admin' from: 10.10.10.10 (Local Database)
```

```
Enter a host name or IP address: 10.10.20.1

PING 10.10.20.1 (10.10.20.1): 56 data bytes
64 bytes from 10.10.20.1: icmp_seq=0 ttl=64 time=1.654 ms
64 bytes from 10.10.20.1: icmp_seq=1 ttl=64 time=0.109 ms
64 bytes from 10.10.20.1: icmp_seq=2 ttl=64 time=0.044 ms

--- 10.10.20.1 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 0.044/0.602/1.654/0.744 ms
```

```
Enter a host name or IP address: 10.10.30.1

PING 10.10.30.1 (10.10.30.1): 56 data bytes
64 bytes from 10.10.30.1: icmp_seq=0 ttl=64 time=0.857 ms
64 bytes from 10.10.30.1: icmp_seq=1 ttl=64 time=0.041 ms
64 bytes from 10.10.30.1: icmp_seq=2 ttl=64 time=0.358 ms

--- 10.10.30.1 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 0.041/0.418/0.857/0.336 ms
```

Phần 2 — Active Directory

Cấu hình DHCP Server cấp phát dải 10.10.10.100 – 10.10.10.200 cho VLAN Client

Interfaces / VLANs

Interface Assignments

Interface Groups

Wireless

VLANs

QinQs

PPPs

GREs

GIFs

Bridges

LAGGs

VLAN Interfaces				
Interface	VLAN tag	Priority	Description	Actions
em1 (lan)	10		SERVER	 
em1 (lan)	20		CLIENT	 

+

 Add

Primary Address Pool

Subnet

10.10.30.0/24

Subnet Range

10.10.30.1 - 10.10.30.254

Address Pool Range

10.10.30.100

From

10.10.30.200

To

The specified range for this pool must not be within the range configured on any other address pool for this interface.

Additional Pools

+

 Add Address Pool

If additional pools of addresses are needed inside of this subnet outside the above range, they may be specified here.

Tạo tối thiểu 2 VLAN: VLAN 10 (Server) và VLAN 20 (Client);

Interfaces / Interface Assignments

Interface Assignments

Interface Groups

Wireless

VLANs

QinQs

PPPs

GREs

GIFs

Bridges

LAGGs

Interface	Network port	
WAN	em0 (00:0c:29:7e:48:76)	
LAN	em1 (00:0c:29:7e:48:80)	 Delete
SERVER	VLAN 10 on em1 - lan (SERVER)	 Delete
CLIENT	VLAN 20 on em1 - lan (CLIENT)	 Delete

Save

cấu hình định tuyến inter-VLAN

Firewall / Rules / SERVER

Floating WAN LAN **SERVER** CLIENT

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	0/0 B	IPv4 TCP	SERVER subnets	*	CLIENT subnets	*	*	none		     

 Add  Add  Delete  Toggle  Copy  Save  Separator

Firewall / Rules / CLIENT

Floating WAN LAN SERVER **CLIENT**

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	0/0 B	IPv4 TCP	CLIENT subnets	*	SERVER subnets	*	*	none	ALLOW CLIENT TO SERVER	     

 Add  Add  Delete  Toggle  Copy  Save  Separator

Viết firewall rule: chỉ cho phép VLAN Client truy cập Web (80/443) và DNS ra ngoài

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	0/0 B	IPv4 TCP	CLIENT subnets	*	*	443 (HTTPS)	*	none		     
☐	✓	0/0 B	IPv4 TCP	CLIENT subnets	*	*	80 (HTTP)	*	none		     

chặn hoàn toàn truy cập trực tiếp từ VLAN Client đến các cổng database (MySQL 3306, PostgreSQL 5432, Redis 6379).

The changes have been applied successfully. The firewall rules are now reloading in the background.
[Monitor](#) the filter reload progress.

IP Ports URLs All

Firewall Aliases Ports

Name	Type	Values	Description	Actions
DATABASE_PORTS	Port(s)	3306, 5432, 6379	BLOCK MYSQL POSTGRESQL REDIS FROM CLIENT	  

 Add  Import

Floating WAN LAN SERVER CLIENT

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	 0/0 B	IPv4 TCP	CLIENT subnets	*	SERVER subnets	DATABASE_PORTS	*	none			     
☐	 0/0 B	IPv4 TCP	CLIENT subnets		*	443 (HTTPS)	*	none			     
☐	 0/0 B	IPv4 TCP	CLIENT subnets	*	*	80 (HTTP)	*	none			     

Cấu hình NAT port-forward để Kibana (5601) và Grafana (3000) truy cập được từ mạng quản trị bên ngoài LAN.

The changes have been applied successfully. The firewall rules are now reloading in the background.
[Monitor](#) the filter reload progress.

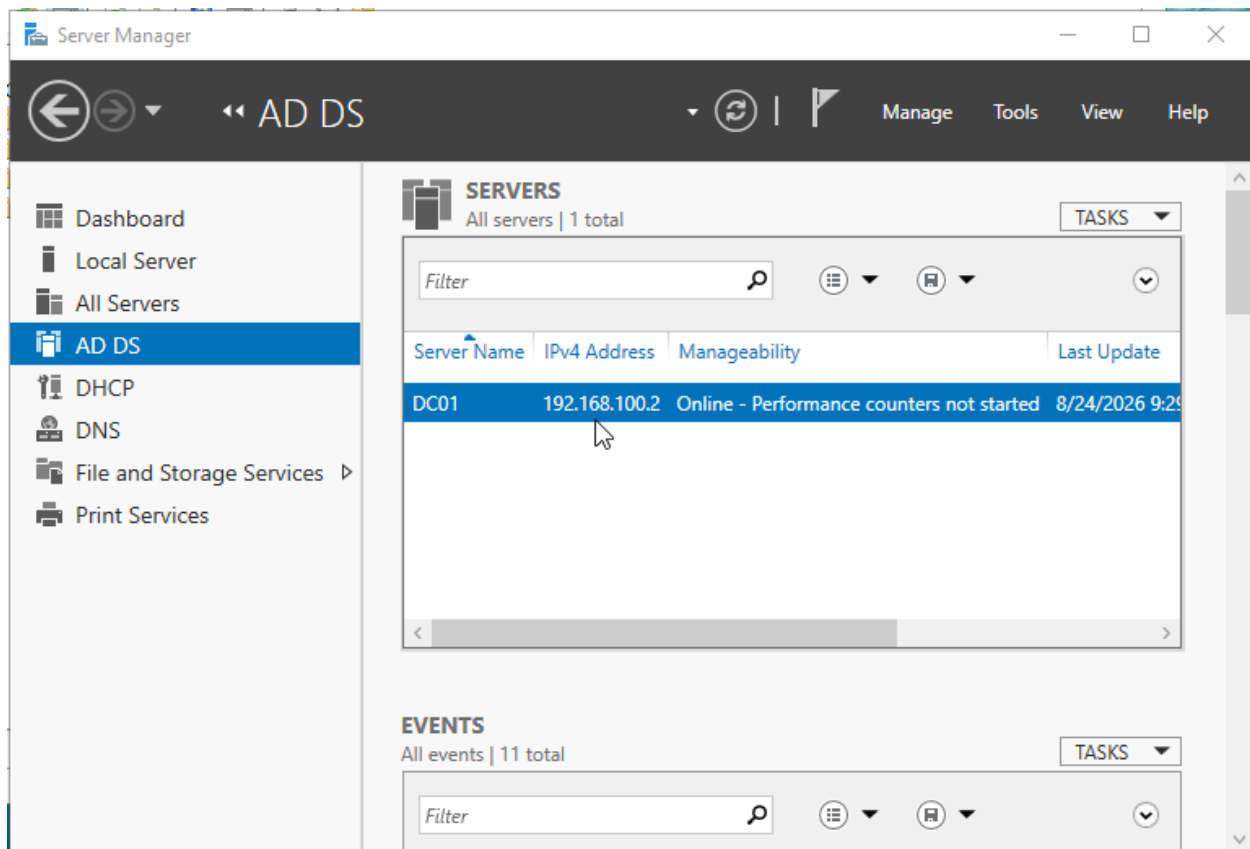
Port Forward 1:1 Outbound NPT

Rules

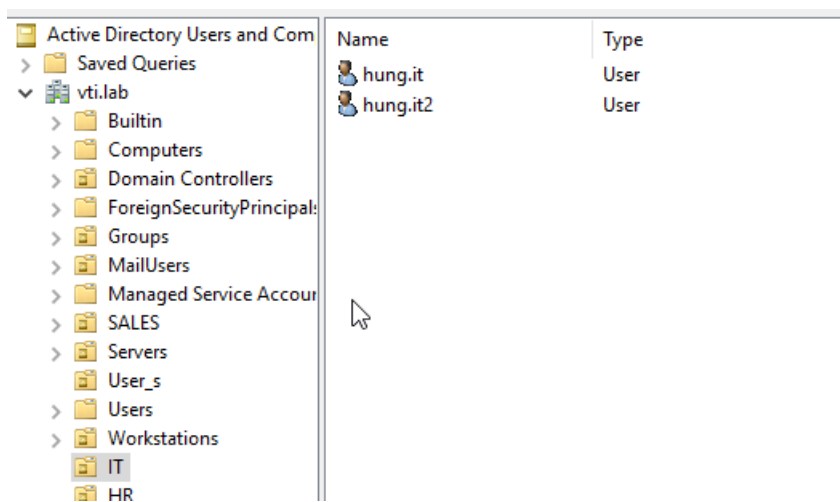
☐	Interface	Protocol	Source Address	Source Ports	Dest. Address	Dest. Ports	NAT IP	NAT Ports	Description	Actions
☐	  WAN	TCP	*	*	WAN address	3000 (HBCI)	10.10.20.11	3000 (HBCI)	NAT_GRAFANA_300	  
☐	  WAN	TCP	*	*	WAN address	5601	10.10.20.10	5601	NAT_KIBANA_5601	  

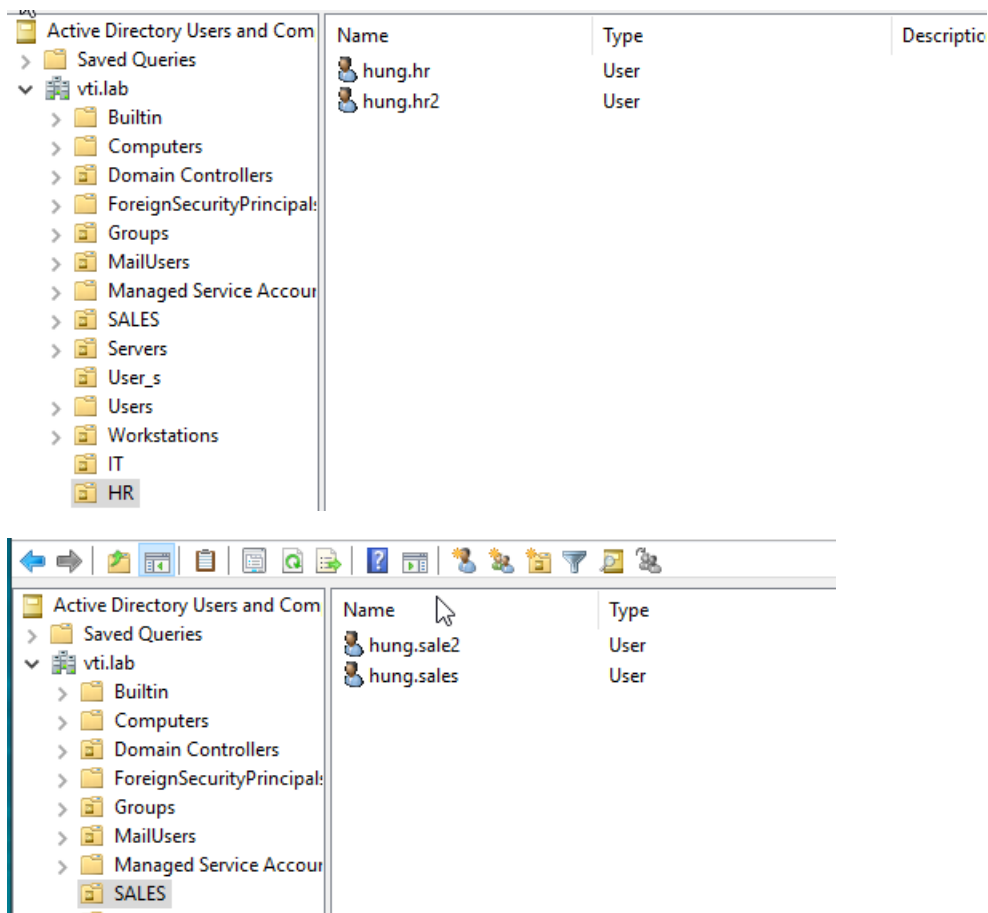
 Add Add Delete Toggle Save Separator

Cài đặt AD DS và promote DC-01 thành Domain Controller cho domain vti.lab

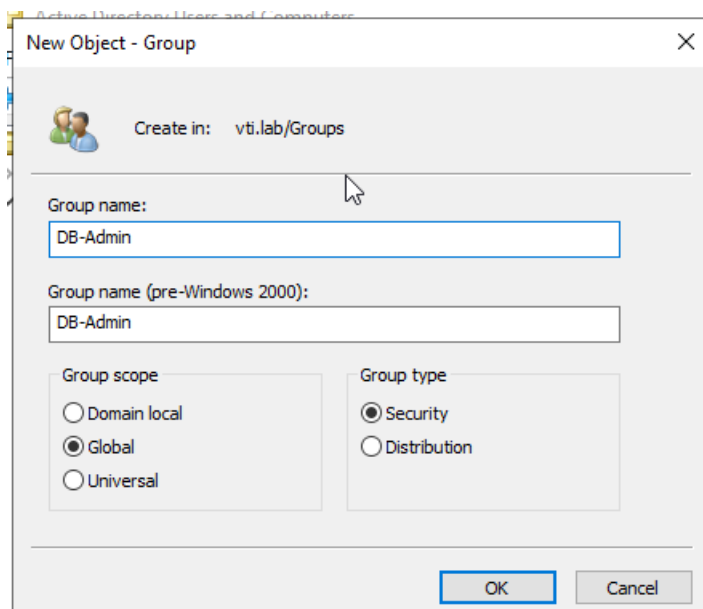


Tạo cấu trúc OU: IT, Sales, HR; tạo tối thiểu 6 user và phân bổ vào các OU tương ứng





Tạo 2 Security Group: DB_Admins và Monitoring_Admins



Monitoring_Admins Properties

General Members Member Of Managed By

 Monitoring_Admins

Group name (pre-Windows 2000):

Description:

E-mail:

Group scope

☐ Domain local

☒ Global

☐ Universal

Group type

☒ Security

☐ Distribution

Notes:

OK Cancel Apply

thêm user phù hợp vào mỗi nhóm

DB-Admin Properties

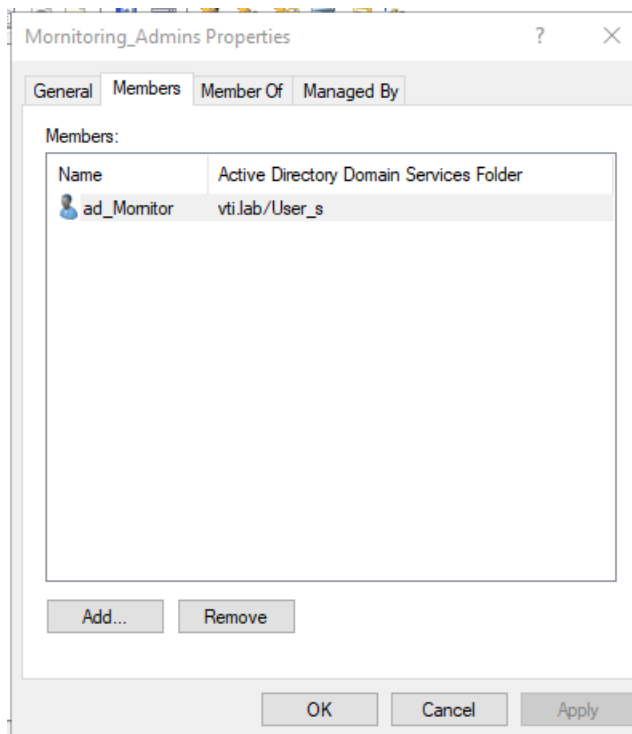
General Members Member Of Managed By

Members:

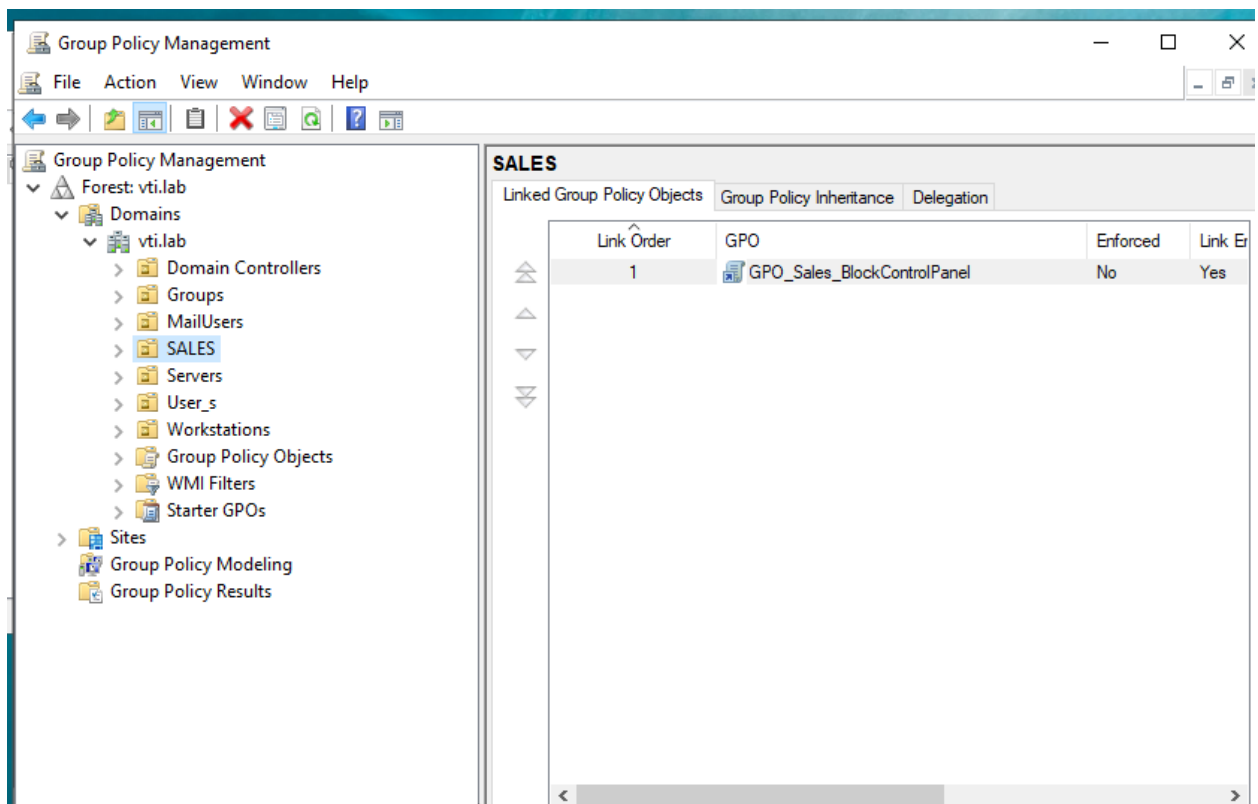
Name	Active Directory Domain Services Folder
 ad_DB	vti.lab/User_s

Add... Remove

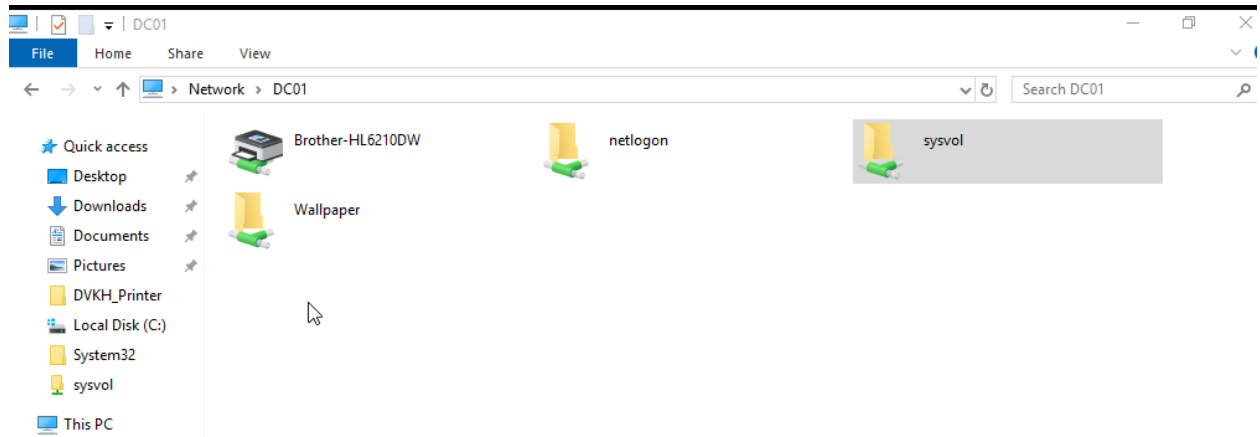
OK Cancel Apply



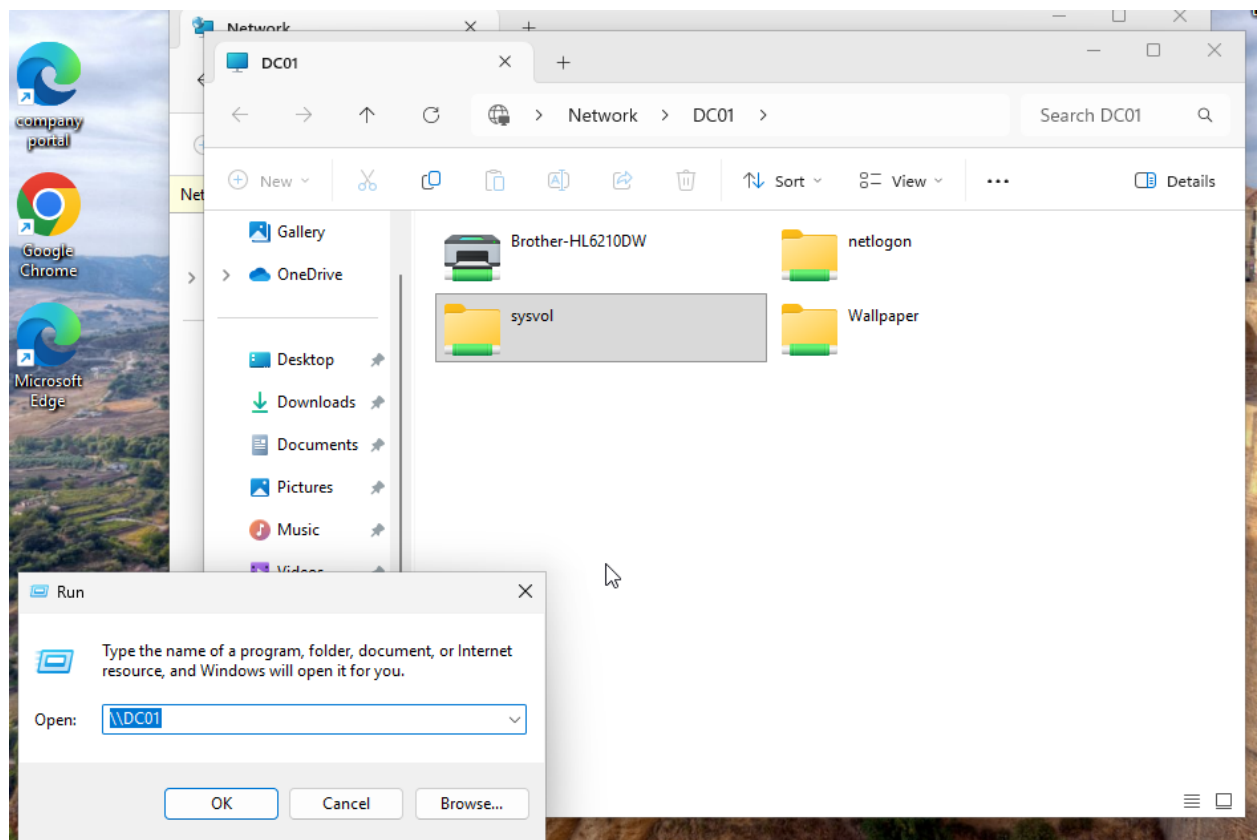
Áp dụng 1 GPO khóa Control Panel cho OU Sales

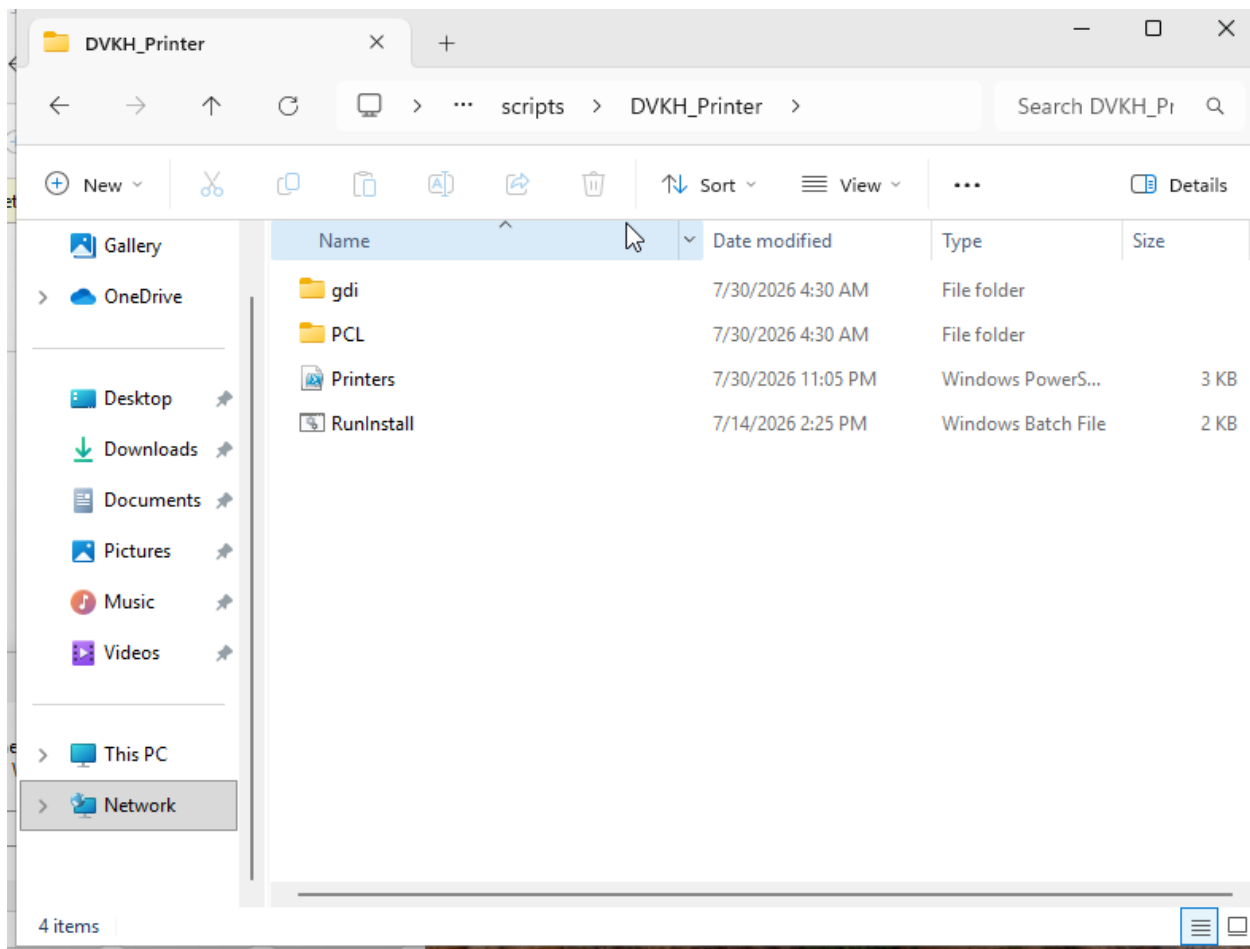


1 GPO map ổ đĩa mạng áp dụng cho toàn domain

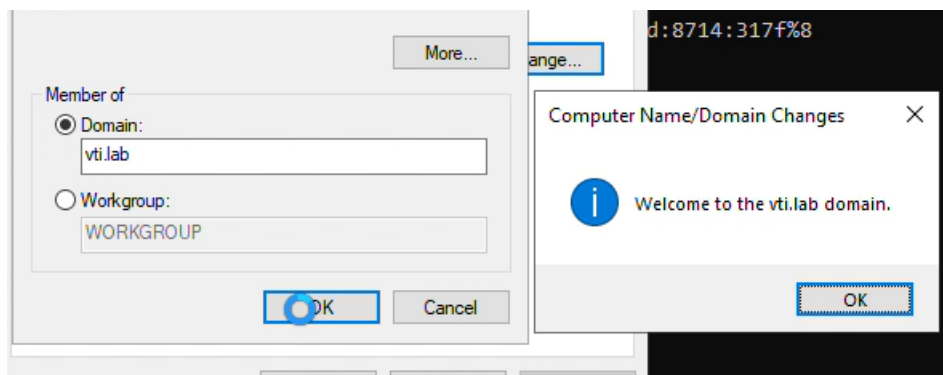


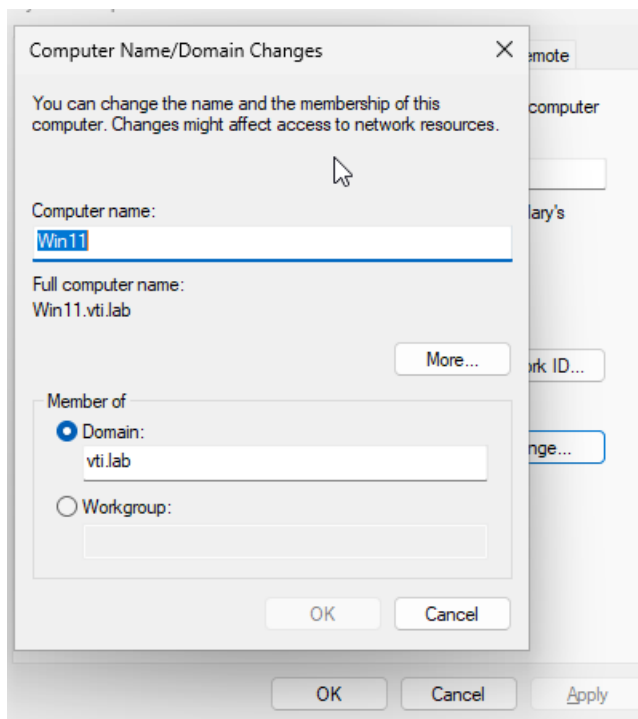
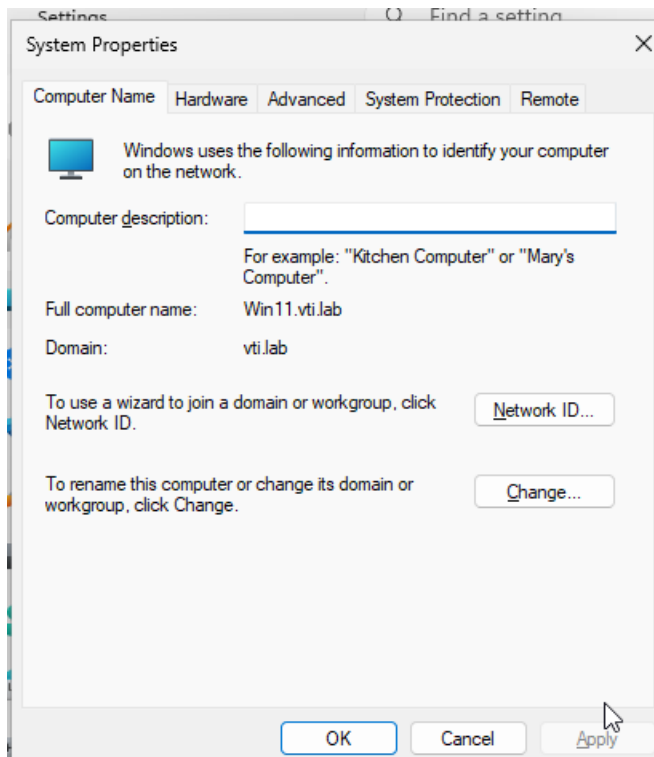
User trong domain có thể truy cập ổ dung chung

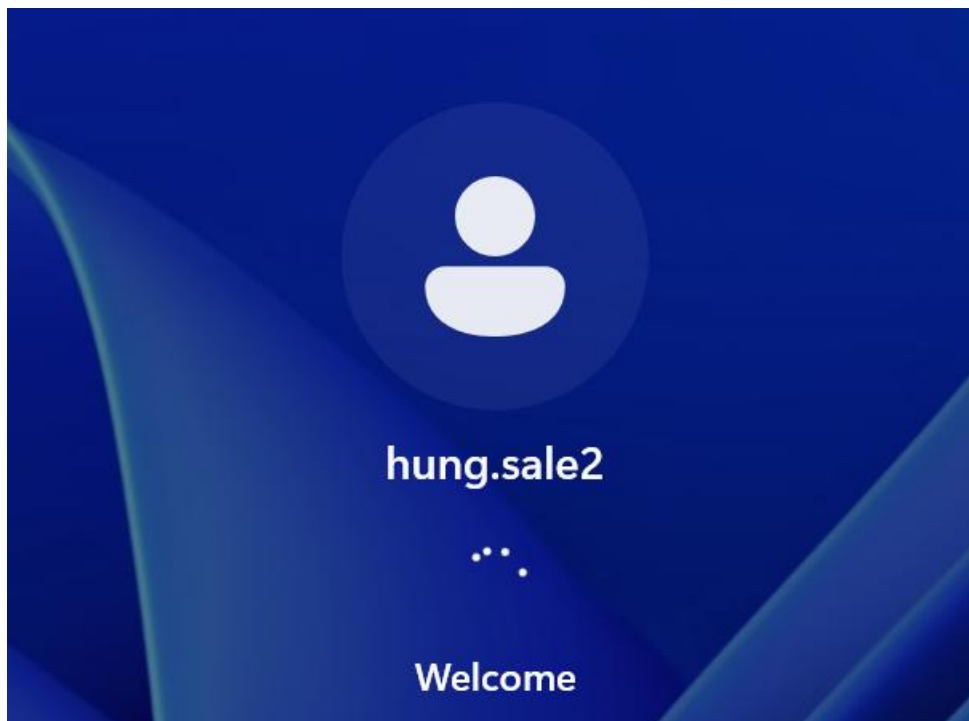




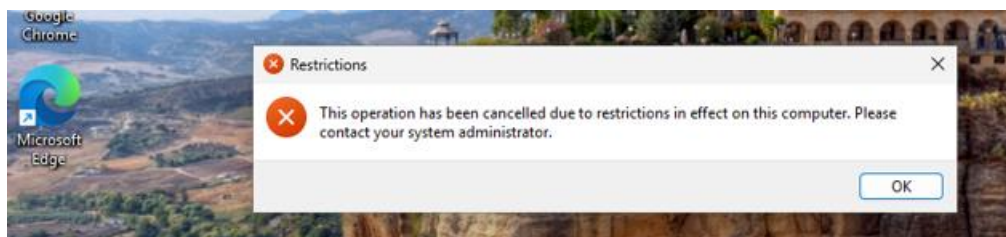
Join máy client vào domain và đăng nhập thành công bằng một user vừa tạo







GPO policy blockcontrol panel chạy trên user sale



GPO không hoạt động trên user khác

